SPOT AND THE SCREWDRIVER

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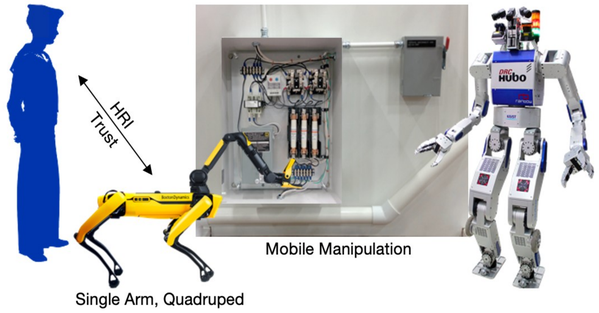
MOTIVATION

Spot is an agile mobile quadrupedal robot developed by Boston Dynamics. My goal was to get Spot to recognize and operate a screwdriver to open an electrical box for inspection. My work over the summer is part of the HRI Lab's ongoing research on bipedal and quadrupedal robot autonomy in complex unstructured environments in collaboration with Brown University, the Office of Naval Research, and the Naval Undersea Warfare Center.
The aim of this project is to increase the impact and use of robotics by the US Navy and DoD by creating a new integrated framework for perception, planning, navigation, and manipulation under uncertainty for humanoid and quadrupedal robots. The objective is to significantly advance the state of the art in autonomous legged robots operating in highly complex unstructured environments in collaboration with humans in order to create a complete system for assisting Marines on patrol or sailors in a vessel, whether for routine maintenance or for emergency situations.[1]



1. "Bipedal and Quadrupedal Robot Autonomy in Complex Unstructured Environments." UML HRI Lab. https://www.uml-hri-lab.com/research#h.5924aanpglrl.

APPROACH

FIND THE BOX
- DETECT LOCATION OF BOX
- DETERMINE ORIENTATION OF SCREWS

GET THE SCREWDRIVER
- PICK UP SCREWDRIVER FROM PERSON/LOCATION
- DETERMINE ORIENTATION OF SCREWDRIVER

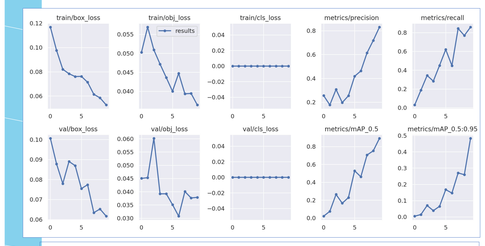
OPERATE ON THE BOX
- NAVIGATE TO BOX USING SAVED LOCATION
- TURN THE SCREWS BASED ON SAVED ORIENTATIONS

I chose this unconventional approach due to a limitation of Spot's arm. Since the gripper camera was located in the "palm" of Spot's gripper, the camera would be obscured by any object that Spot would pick up (like a screwdriver).

METHODS

AR Markers

- For identifying and locating the box that we want to open.
- Useful for detecting large objects in noisy environments.
- I used the ARUCO library for the AR markers but will most likely switch to AprilTags for consistency with Spot's fiducials.



Results of training a YOLOv5 model over the image set.

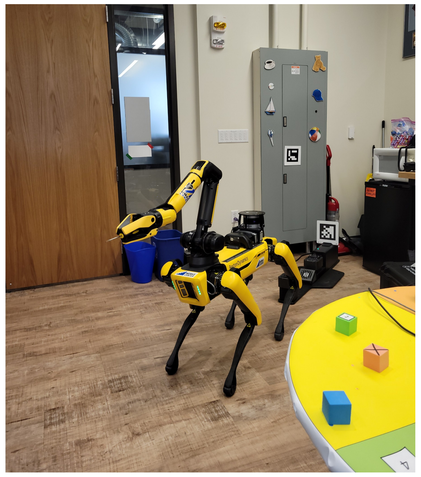
Machine Learning

- For detecting and locating the screwdriver in noisy environments.
- Trained a YOLOv5 model with Pytorch over a set of images with over 800 screwdriver instances.
- I tried various models but chose the YOLOv5 model for its ability to detect small and semi-obscured objects in noisy environments.

Predefined Poses for Grasping and Manipulation

- Approaching and analyzing the box.
- Approaching and grasping the screwdriver.
- Analyzing the orientation of the screwdriver.
- Having predefined poses reduces the amount of computation needed to complete a task and improves consistency.

(Right) Spot in the "viewing" pose after grasping the screwdriver.

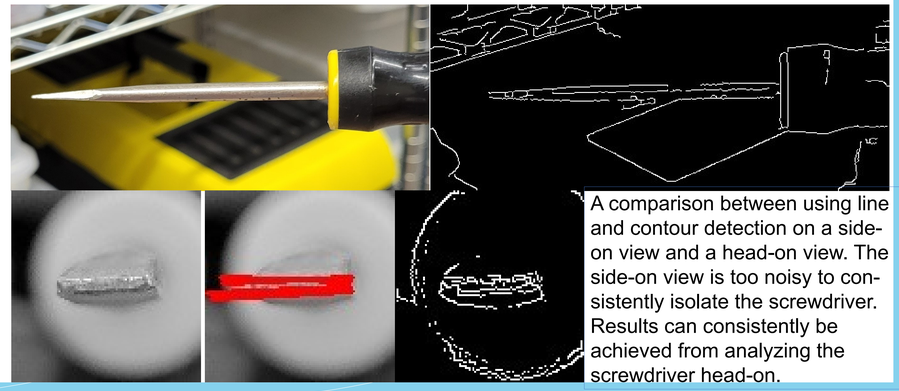


(Above) Spot in the "carry" pose after grasping the screwdriver.

OpenCV for Line and Contour Detection

- Used canny edge detection and contour detection to isolate the screws.
- Used a Haar Cascade Classifier to isolate the screwdriver.
- Used Probabilistic Hough Line Detection (and some trigonometry) to determine the angles of the screws and screwdriver.

Line and contour detection to determine screw orientation.



A comparison between using line and contour detection on a side-on view and a head-on view. The side-on view is too noisy to consistently isolate the screwdriver. Results can consistently be achieved from analyzing the screwdriver head-on.

RESULTS

Trained a detection model for screwdrivers with ~90% accuracy.*

Spot can determine the orientation of a screwdriver that is given to it.

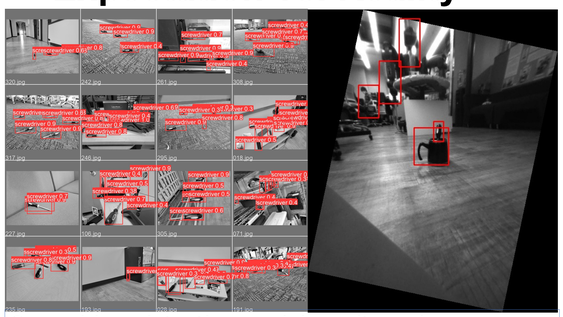
Many of the computer vision modules that I developed for Spot produce fairly consistent results.

CONCLUSIONS

*90% accuracy in the training environment does not translate to 90% accuracy in the testing environment.

Traditional computer vision methods do well in controlled environments while machine learning methods fare better in noisy environments.

Expectation vs. Reality



Screwdriver detection model results on images similar to those in its training set (left). Same model on Spot's fisheye lens (right).

FUTURE WORK

I plan on coming back to continue my work with the HRI lab in the Fall semester.

Further testing and improvement needs to be done on the screwdriver detection and grasping module.

Further testing and improvement needs to be done on the box detection and navigation module.

Further testing and improvement needs to be done on the screw analysis module.

I will most likely retrain the YOLOv5 model with additional images from Spot's cameras.

A module for operating on the box still needs to be developed.

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